

Socioeconomic deprivation and post-discharge mortality in heart failure: evidence from a nationwide universal health coverage system

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Aims

Heart failure (HF) is characterized by high mortality rates during hospitalization and after discharge. Socioeconomic deprivation and inequalities are often barriers to effective treatment, substantially impacting mortality in chronic illnesses. We investigated the association between socioeconomic deprivation, all-cause mortality, and HF readmission among patients hospitalized for HF in France.

Methods and results

We conducted a retrospective cohort study including all adults hospitalized for worsening HF in France in 2018, using data from the national health insurance database. We estimated the association between area-level socioeconomic disadvantage quintiles and income inequality quantified by the Gini index and the outcomes of all-cause mortality and HF readmission using adjusted survival models. Among 92 111 patients (median age 82 years; 50.5% female), 29.7% were readmitted and 25.9% died within 1 year. Higher levels of deprivation were associated with increased one-year mortality, with HR ranging from 1.03 to 1.06, but not consistently across all quintiles. This association persisted in adjusted analysis at 1 (HR 1.11, 1.07–1.16, $P < 0.001$, for the most vs. least deprived) and 2 years (HR 1.11, 1.07–1.15, $P < 0.001$). In contrast, socioeconomic deprivation was not significantly associated with readmission risk, and the Gini index showed no association with either mortality or readmission.

Conclusion

In a country with universal healthcare, area-level socioeconomic deprivation was modestly associated with higher mortality after HF hospitalization, but not with readmission risk. Society-level income inequality showed no relationship with either outcome.

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Graphical abstract

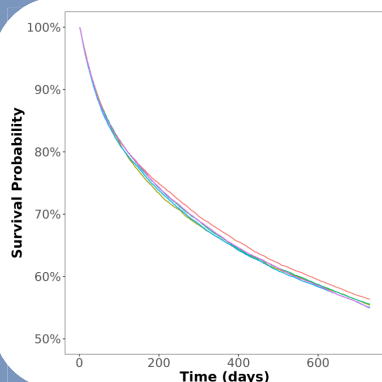
Socio-Economic Deprivation and Post-Discharge Mortality in Heart Failure: Evidence from a Nationwide Universal Health Coverage System

Background: Socioeconomic deprivation may worsen outcomes for HF patient

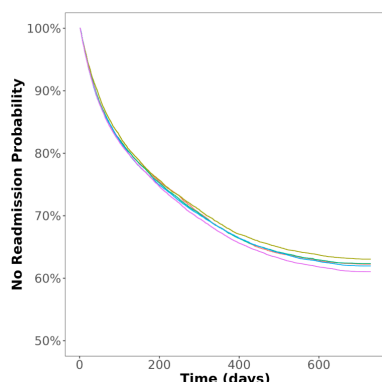
- **Cohort:** 92,111 French HF patients hospitalized in 2018

- **Exposures:** Deprivation & Gini index

- **Outcomes:** Mortality and HF readmission at 1 & 2 years



Overall Survival by Deprivation Quintiles



Readmission by Deprivation Quintiles

Deprivation Quintiles
— 1
— 2
— 3
— 4
— 5

Results :

Deprivation: ↑ 11% mortality at 1 & 2 years between Q1 and Q5

Readmission: No association with deprivation

Gini index: No link to mortality or readmission

Socioeconomic gradient in HF prevalence → **Gaps in primary prevention**
 No effect on readmission → **Equitable in universal healthcare access once hospitalized**
 France (~30% readmission) vs USA (~63% reported) → **Healthcare systems strongly shape outcomes**
 Deprivation +11% mortality risk at 1 and 2 year → **Modest association, buffering effect of universal healthcare**

Keywords

Heart failure • Socioeconomic factors • Mortality • Hospital readmission

Key Learning Points

What is already known:

- Socioeconomic deprivation is a recognized risk factor for poor heart failure outcomes.
- Evidence mainly comes from US studies, where healthcare access inequalities strongly shape these disparities.

What this study adds:

- In France's universal healthcare system, deprivation was linked to only modestly higher mortality and not to rehospitalization.
- Income inequality (Gini index) showed no impact on mortality or rehospitalization.
- Universal coverage may buffer socioeconomic disparities after HF hospitalization, but disadvantaged groups remain over-represented among patients.

Introduction

Heart failure (HF) is characterized by episodes of acute decompensation and high mortality rates, not only during hospitalization,^{1,2} but also in the post-discharge period.³ One-year mortality after hospital discharge for HF remains high worldwide, with rates typically ranging from 13% to over 36% depending on population and region.³⁻⁸ In 2022, there were more than 180 000 hospitalizations for HF in France, with 17% of these patients being readmitted within 6 months.^{9,10} Consequently, patients with a history of hospitalization for HF represent a significantly large and vulnerable population.

It is crucial to understand the factors influencing outcomes in at-risk patients with prior hospitalizations—not only to enhance diagnosis and treatment, but also to reduce mortality.

Socioeconomic deprivation refers to a multi-dimensional lack of resources, opportunities, and social participation and is distinct from the concept of poverty.¹¹ It is measured using indicators (e.g. income, education, employment, housing, and social relationship) at both the individual and area levels and is a key determinant of health inequalities.^{12,13} Socioeconomic deprivation poses substantial barriers to accessing effective care and achieving optimal treatment outcomes.¹⁴⁻¹⁷ In the context of HF, it can become a major risk factor for both rehospitalization or mortality.¹⁸ In

Taiwan, based on the National Health Insurance Research Database, a 29% higher risk of hospital readmission and nearly a twofold increase in all-cause mortality were observed among low-income patients with HF compared with high-income individuals.¹⁹ This highlights an uneven distribution of the burden of HF across socioeconomic groups, with disparities in care and outcomes including mortality rates. Findings are not uniform across studies, as detailed analyses of context, measured outcomes, and socioeconomic status (SES) indicators reveal significant variations.^{7,20} Although France is often regarded as a robust model of near-universal health coverage, we hypothesized that socioeconomic inequalities may exert a substantial influence on both access to care and clinical outcomes.

To that purpose, we aimed to investigate whether socioeconomic deprivation and income inequality, as measured by the area-level deprivation index and GINI index—a measure of income inequality—respectively, are associated with mortality and HF readmission after an index hospitalization for HF.

Methods

Study design and population

This observational retrospective nationwide cohort study focused on adults hospitalized for HF decompensation in France in 2018. Eligible patients were French adults enrolled in the mandatory national health insurance programme, with at least one hospital admission in 2018, primarily for HF (ICD-10 code: I50). Subjects were required to have spent at least one night in the hospital, with admission from and discharge to their residence.

Data source

Data were collected from the SNDS (*Système national des données de santé*), the medical-administrative database covering the entire French population.²¹ The research utilized algorithms for retrieving comorbidity within the disease mapping system developed by the French national health insurance agency.²² We employed an exhaustive methodology, including all patients without missing data meeting the inclusion criteria. For the exposure variables, the deprivation index was directly available in the SNDS, where it is routinely provided at the municipal level as a standardized indicator. In contrast, the Gini index is not included in the SNDS and was therefore obtained from publicly accessible datasets produced by the French National Institute of Statistics and Economic Studies, which publishes harmonized income-distribution measures for all municipalities.²³ We selected 2018 as the index year to ensure complete follow-up unaffected by the COVID-19 pandemic, as subsequent years would have introduced major disruptions in care pathways and mortality that could bias the interpretation of socioeconomic associations.

Study variables

Data retrieved included sociodemographic factors (age, sex, deprivation, and Gini index at area level), comorbidities from the past 5 years (ischemic heart diseases, cerebrovascular diseases, peripheral vascular diseases, diabetes, cancer, dementia, Alzheimer's disease, chronic respiratory diseases, and end-stage renal diseases), health-care utilization (general practitioner and cardiologist visits, nursing care, treatment, and pro-natriuretic peptide tests), characteristics of the index hospitalization [duration, admission in intensive care unit (ICU), and status of the establishment], readmission rates, and patient outcomes of survival after 1 and 2 years.

The deprivation index score utilized in this study was the well-established French deprivation index, an area-level variable calculated at the municipal level. The municipal level is the smallest administrative divisions in France, totalling 30 500 units, with a median population of 500. This score is based on four socioeconomic factors: median income, high school graduation rates, workforce participation, and unemployment rates.²⁴ This score was categorized

into quintiles, with Q1 representing the least disadvantaged group and Q5 representing the most disadvantaged, as defined for the general population. The deprivation index was assigned based on the patient's residential municipality. Patients will be described according to these quintiles (Q1–Q5) to ensure comparability with the general population. Consequently, the distribution of quintiles is not expected to be balanced within the patient population.

In addition to the deprivation index, the study used the Gini index as a measure of income inequality at area level. The Gini index ranges from 0 to 1, where 0 signifies perfect equality (everyone has the same income) and 1 indicates the maximum inequality.²⁵ In the absence of data due to statistical confidentiality for municipalities with fewer than 1000 residents, we used the Gini index at the municipal grouping level. We imputed the remaining 1.7% of missing data by the mean.

Additionally, the study employed the HF stability score²⁶ to classify patients into four groups based on their hospitalization history and long-term disease (LTD) progression. This score ranges from 1 to 4, defined by the following criteria: (1) no HF hospitalization/or LTDs in the past 36 months; (2) no HF hospitalization, but LTD present; (3) one HF hospitalization in the past 12 months; and (4) ≥ 2 HF hospitalizations in the past 12 months.

The database distinguishes scheduled HF management visits from unplanned acute readmissions. Only unplanned admissions for acute HF decompensation were classified as readmission events.

The primary outcome of this study was all-cause mortality among French patients hospitalized for HF at 1 year. All-cause mortality was chosen because it is the most objective and consistently recorded outcome in administrative datasets. Secondary outcomes included all-cause mortality at 2 years and rehospitalization for HF at 1 and 2 years.

Statistical methods

The descriptive analysis was conducted on the entire population and within deprivation index quintile groups. For quantitative variables, the results are presented as means $\pm$ standard deviations and median with first and third quartiles. For categorical variables, frequencies and corresponding percentages are reported for each category. Comparisons between groups were made using the comparison of means tests (Student or Wilcoxon, depending on distribution) for quantitative variables and a chi-square test for qualitative variables. When the validity conditions of the chi-square test were not met, it was replaced by Fisher's exact test.

Effect size was utilized to illustrate the differences between the first and last quintiles of deprivation. Cohen's *d* was calculated for quantitative variables, while Cohen's *h* was applied for qualitative variables. Cohen's *d* and *h* effect sizes are interpreted as 0.2 (small), 0.5 (medium), and 0.8 (large) using established benchmarks.²⁷

Univariable and multivariable Cox regression analyses were conducted to evaluate the relationship between socioeconomic deprivation and all-cause mortality at 1 and 2 years. The multivariable analysis included deprivation and the Gini index at area level and was adjusted on previously cited factors: age, sex, comorbidities from the past 5 years, HF stability score, general practitioner and cardiologist visits, nursing care, treatment and pro-natriuretic peptide tests, and characteristics of the index hospitalization. To model the occurrence of HF readmission, we employed multivariable competing risks models to account for the risk of death, while also adjusting for previous factors. The proportional hazards assumption was assessed using Schoenfeld residuals. Statistical significance was set at a *P*-value of 0.05 for two-sided tests. The reporting of this observational study complies with the STROBE recommendations. In accordance with French law and the CNIL framework, this retrospective observational study did not require formal ethical approval, as patient anonymity was fully preserved.

Results

Population

The final cohort included 92 111 patients, with a median age of 82 years (interquartile range: 73; 88). Half (50.5%) of the

population was female. At the one-year mark, 25.9% of participants had died, with an average time to death of 144 days. The effect sizes (Cohen's *d* and *h*) did not exceed 0.22 between the least disadvantaged (Q1) and the most disadvantaged quintiles (Q5) for any of the variables. Sex distribution remained consistent across the quintiles, with approximately half of the patients in each quintile being female. Age distribution was also even between groups. The number of patients increased across the quintiles, with Q1 representing 16% and Q5 rising to 25%. Additionally, the mean length of stay increased from 8.59 days in Q1 to 9.00 days in Q5. It is also important to note that quintiles were defined based on the general population. See [Table 1](#).

Outcome

The one-year readmission rate was 29.7%, and the mortality rate was 26%. The readmission rates ranged from 28.8% to 30.4% between quintiles, and the mortality rates ranged from 25.1% to 26.5%. At 2 years, the readmission rate was 33.8% and the mortality rate was 39.6%. The two-year readmission rates varied between 32.8% and 34.8% between quintiles, and the two-year mortality rates ranged from 38.6% to 40.0%. See [Table 2](#).

Association of deprivation index with all-cause mortality

The univariable analysis indicated significant relationships between the deprivation index quintiles and the one-year risk of mortality (see [Figure 1](#)). Q1 was the reference group. Q2 presented a crude hazard ratio (HR) of 1.06 (95% CI: 1.02–1.11, $P = 0.007$). Patients in Q3 had a crude HR of 1.04 (95% CI: 0.99–1.08, $P = 0.1$). Those in Q4 exhibited a crude HR of 1.05 (95% CI: 1.01–1.09, $P = 0.03$). Lastly, the most disadvantaged group Q5 had a crude HR of 1.03 (95% CI: 0.99–1.07, $P = 0.2$). We also tested for a potential clustering effect at the municipality level using a shared frailty model. The HR estimates were identical to those from the standard Cox model. The frailty variance was negligible (0.0034), justifying the use of a standard Cox model.

In the adjusted analysis—accounting for sociodemographic factors, comorbidities over the past 5 years, healthcare utilization, and index hospitalization characteristics—only the most advantaged group (Q1) differed from the others: individuals in Q2–Q5 all had a similarly higher risk of mortality (adjusted HR ranging from 1.07 to 1.11, with all $P < 0.01$).

Similar results were found at the two-year mark, with Q2 exhibiting an adjusted HR of 1.07 (95% CI: 1.03–1.11, $P = <0.001$), Q3 at 1.05 (95% CI: 1.02–1.09, $P = 0.004$), Q4 at 1.09 (95% CI: 1.05–1.13, $P = <0.001$), and Q5 at 1.11 (95% CI: 1.07–1.15, $P = <0.001$). Detailed results are presented in [Table 3](#).

Association of deprivation index with readmission

Socioeconomic disadvantage was not significantly linked to an increased risk of HF readmission at either 1 or 2 years (see [Figure 2](#)). At 1 year, HR ranged from 0.98 (Q2) to 1.03 (Q5), with all P -values greater than 0.10. At 2 years, HRs remained consistent, ranging from 0.98 (Q2) to 1.04 (Q5), with all P -values exceeding 0.06. These findings indicated that socioeconomic disadvantage did not demonstrate a statistically significant association with HF readmission risk at either 1 or 2 years. Details can be seen in [Table 4](#).

Association of Gini index with all-cause mortality or readmission

Notably, the GINI index showed no significant association with overall mortality ($P = 0.44$ at 1 year and $P = 0.49$ at 2 years) or readmission rates ($P = 0.42$ and $P = 0.78$) after hospitalization.

Discussion

Main findings

This study examined the relationship between area-level socioeconomic deprivation and outcomes after hospitalization for HF in France, with a particular focus on the role of socioeconomic factors in mortality and readmission. Using a comprehensive nationwide cohort of adults hospitalized for HF decompensation, we found limited support for our initial hypothesis. After adjustment for sociodemographic characteristics, comorbidities, healthcare utilization, and index hospitalization factors, socioeconomic deprivation showed only a modest association with mortality: patients in the most disadvantaged quintile had an approximately 11% higher one-year and two-year mortality risk compared with the least disadvantaged group, while risks were similar across quintiles Q2–Q5. By contrast, socioeconomic deprivation was not significantly associated with HF readmission at either 1 or 2 years, and the Gini index was unrelated to both mortality and readmission.

Socioeconomic gradient in HF prevalence

The progressively higher proportion of patients across deprivation quintiles—from 16% in Q1 to 25% in Q5—indicates that individuals from more disadvantaged areas are over-represented among those hospitalized for HF. This pattern reflects a socioeconomic gradient in HF prevalence, consistent with previous literature linking economic deprivation to a greater burden of comorbidities and HF.²⁰ This socioeconomic gradient in the prevalence of HF indicates that, in France, as in many other countries, disadvantaged populations are disproportionately affected by this condition, revealing notable shortcomings in primary prevention efforts.

Remarkably, most groups' characteristics, such as age, gender, and comorbidities, showed a minimal variation between quintiles. This is especially significant considering that HF is a complex and heterogeneous condition, where patient characteristics and underlying cardiac function frequently vary, indicating the existence of distinct subgroups within this population.²⁸

Association with deprivation

The burden of deprivation was less severe than initially hypothesized. Although the deprivation index was identified as a predictor of one- and two-year mortality, its relationship seems limited, with only an 11% increase in mortality risk between the least disadvantaged and the most disadvantaged quintiles. The increased risk was consistent across the last four quintiles, ranging from 5% to 11%. This takes place within the egalitarian framework of the French healthcare system, which ensures universal access to medical care with minimal financial barriers. Despite the well-established link between socioeconomic disadvantage and poorer health outcomes, these structural features may help to attenuate some of the resulting health inequalities. There is also no clear evidence linking socioeconomic disadvantage to an increased risk of HF readmission in this population, at neither 1 nor 2 years. This suggests that while our healthcare

Table 1 Cohort characteristics

Variable	Modality	Population, n = 92 111	Deprivation index quintile					Effect size (Cohen's d and Cohen's h) and 95% CI between Q1 and Q5
			Q1, n = 14 673 (16)	Q2, n = 15 892 (17)	Q3, n = 18 596 (20)	Q4, n = 20 156 (22)	Q5, n = 22 794 (25)	
Sex, n (%)	Female	46 477 (50.46)	7320 (49.89)	7889 (49.64)	9354 (50.30)	10 111 (50.16)	11 803 (51.78)	-0.04 (-0.06 ; -0.02)
Age group, n (%)	18-64	10 027 (10.89)	1475 (10.05)	1713 (10.78)	2006 (10.79)	2174 (10.79)	2659 (11.67)	-0.05 (-0.07 ; -0.03)
	65-74	16 596 (18.02)	2455 (16.73)	2797 (17.60)	3318 (17.84)	3695 (18.33)	4331 (19.00)	-0.06 (-0.08 ; -0.04)
	75-84	28 494 (30.93)	4407 (30.03)	4838 (30.44)	5802 (31.20)	6237 (30.94)	7210 (31.63)	-0.03 (-0.06 ; -0.01)
	85-94	32 598 (35.39)	5534 (37.72)	5737 (36.10)	6545 (35.20)	7129 (35.37)	7653 (33.57)	0.09 (0.07 ; 0.11)
Establishment status, n (%)	95+	4396 (4.77)	802 (5.47)	807 (5.08)	925 (4.97)	921 (4.57)	941 (4.13)	0.06 (0.04 ; 0.08)
	Private hospital	16 086	3389	3397	3009	3009	3282	0.22 (0.2 ; 0.25)
Length of stay of the index hospitalization (days)	Mean (SD)	8.71 (± 6.87)	8.59 (± 7.76)	8.41 (± 6.60)	8.56 (± 6.98)	8.83 (± 6.53)	9.00 (± 6.61)	-0.06 (-0.08 ; -0.04)
	Median (Q1;Q3)	7.00 (4.00 ; 11.00)	7.00 (4.00 ; 11.00)	7.00 (4.00 ; 11.00)	7.00 (4.00 ; 11.00)	7.00 (5.00 ; 11.00)	8.00 (5.00 ; 11.00)	
Transfer to intensive care, n (%)	Yes	22 838 (24.79)	4065 (27.70)	4058 (25.53)	4318 (23.22)	4707 (23.35)	5690 (24.96)	0.06 (0.04 ; 0.08)
	No							
Severity, n (%)	1: No HF hospitalization/or long-term diseases (LTDs) in the past 36 months	57 455 (62.38)	9148 (62.35)	10 015 (63.02)	11 665 (62.73)	12 501 (62.02)	14 126 (61.97)	0.01 (-0.01 ; 0.03)
	2: No HF hospitalization, but LTD present	8528 (9.26)	1263 (8.61)	1455 (9.16)	1751 (9.42)	1919 (9.52)	2140 (9.39)	-0.03 (-0.05 ; -0.01)
	3: 1 HF hospitalization in 12 months	21 608 (23.46)	3434 (23.40)	3654 (22.99)	4350 (23.39)	4750 (23.57)	5420 (23.78)	-0.01 (-0.03 ; 0.01)
	4: ≥2 HF hospitalizations in 12 months	4520 (4.91)	828 (5.64)	768 (4.83)	830 (4.46)	986 (4.89)	1108 (4.86)	0.03 (0.01 ; 0.06)
Ischemic heart diseases, n (%)	Yes	42 324 (45.95)	6722 (45.81)	7300 (45.94)	8654 (46.54)	9177 (45.53)	10 471 (45.94)	0 (-0.02 ; 0.02)
	No							
Cerebrovascular diseases, n (%)	Yes	11 609 (12.60)	2043 (13.92)	2050 (12.90)	2365 (12.72)	2399 (11.90)	2752 (12.07)	0.06 (0.03 ; 0.08)
	No							
Peripheral vascular diseases, n (%)	Yes	13 683 (14.85)	2051 (13.98)	2451 (15.42)	2844 (15.29)	3005 (14.91)	3332 (14.62)	-0.02 (-0.04 ; 0)
	No							
Diabetes, n (%)	Yes	31 802 (34.53)	4396 (29.96)	5307 (33.39)	6404 (34.44)	7068 (35.07)	8627 (37.85)	-0.17 (-0.19 ; -0.15)
	No							
Cancer, n (%)	Yes	9043 (9.82)	1578 (10.75)	1644 (10.34)	1869 (10.05)	1963 (9.74)	1989 (8.73)	0.07 (0.05 ; 0.09)
	No							
Schizophrenia and delusional diseases, n (%)	Yes	1135 (1.23)	204 (1.39)	192 (1.21)	271 (1.46)	227 (1.13)	241 (1.06)	0.03 (0.01 ; 0.05)
	No							
Depression and mood disorders, n (%)	Yes	4606 (5.00)	791 (5.39)	811 (5.10)	1007 (5.42)	956 (4.74)	1041 (4.57)	0.04 (0.02 ; 0.06)
	No							

Continued

Table 1 Continued

Variable	Modality	Population, n = 92 111	Deprivation index quintile					Effect size (Cohen's d and Cohen's h) and 95% CI between Q1 and Q5
			Q1, n = 14 673 (16)	Q2, n = 15 892 (17)	Q3, n = 18 596 (20)	Q4, n = 20 156 (22)	Q5, n = 22 794 (25)	
Substance abuse disorders, n (%)	Yes	2315 (2.51)	343 (2.34)	372 (2.34)	506 (2.72)	504 (2.50)	590 (2.59)	-0.02 (-0.04 ; 0)
Dementia (including Alzheimer's disease), n (%)	Yes	12 342 (13.40)	2379 (16.21)	2216 (13.94)	2382 (12.81)	2579 (12.80)	2786 (12.22)	0.11 (0.09 ; 0.14)
Parkinson's disease, n (%)	Yes	2403 (2.61)	358 (2.44)	420 (2.64)	520 (2.80)	510 (2.53)	595 (2.61)	-0.01 (-0.03 ; 0.01)
Multiple sclerosis, paraplegia, or tetraplegia, n (%)	Yes	443 (0.48)	84 (0.57)	86 (0.54)	80 (0.43)	76 (0.38)	117 (0.51)	0.01 (-0.01 ; 0.03)
Epilepsy, n (%)	Yes	1235 (1.34)	229 (1.56)	199 (1.25)	257 (1.38)	257 (1.28)	293 (1.29)	0.02 (0 ; 0.04)
Chronic respiratory diseases (including asthma and COPD), n (%)	Yes	30 865 (33.51)	4534 (30.90)	5194 (32.68)	6138 (33.01)	6855 (34.01)	8144 (35.73)	-0.1 (-0.12 ; -0.08)
Rheumatoid arthritis or systemic and connective tissue diseases, n (%)	Yes	3196 (3.47)	558 (3.80)	599 (3.77)	585 (3.15)	692 (3.43)	762 (3.34)	0.02 (0 ; 0.05)
HIV infection or AIDS, n (%)	Yes	203 (0.22)	59 (0.40)	33 (0.21)	45 (0.24)	27 (0.13)	39 (0.17)	0.04 (0.02 ; 0.06)
End-stage renal diseases, n (%)	Yes	2186 (2.37)	366 (2.49)	403 (2.54)	447 (2.40)	449 (2.23)	521 (2.29)	0.01 (-0.01 ; 0.03)
Liver and pancreas diseases (including failures), n (%)	Yes	6321 (6.86)	1084 (7.39)	1115 (7.02)	1220 (6.56)	1299 (6.44)	1603 (7.03)	0.01 (-0.01 ; 0.03)
Number of consultations with a GP in the year preceding the index hospitalization	Mean (SD)	10.75 (± 8.02)	9.64 (± 7.91)	11.20 (± 8.37)	11.07 (± 8.44)	10.79 (± 7.70)	10.84 (± 7.72)	-0.15 (-0.17 ; -0.13)
Number of nursing care sessions in the year preceding the index hospitalization	Median (Q1;Q3) Mean (SD)	9.00 (6.00 ; 14.00) 70.25 (± 115.97)	8.00 (4.00 ; 13.00) 60.12 (± 108.23)	9.00 (6.00 ; 15.00) 71.02 (± 116.28)	9.00 (6.00 ; 15.00) 74.80 (± 119.13)	9.00 (6.00 ; 14.00) 70.57 (± 115.87)	9.00 (6.00 ; 14.00) 72.25 (± 117.65)	-0.11 (-0.13 ; -0.09)
Median (Q1;Q3)		11.00 (1.00 ; 67.00)	7.00 (1.00 ; 47.00)	11.00 (1.00 ; 70.00)	12.00 (1.00 ; 1.00)	11.00 (2.00 ; 68.00)	12.00 (2.00 ; 69.00)	

Continued

Table 1 Continued

Variable	Modality	Population, n = 92 111	Deprivation index quintile					Effect size (Cohen's d and Cohen's h and 95% CI between Q1 and Q5
			Q1, n = 14 673 (16)	Q2, n = 15 892 (17)	Q3, n = 18 596 (20)	Q4, n = 20 156 (22)	Q5, n = 22 794 (25)	
At least one consultation with a cardiologist in the year preceding the index hospitalization, n (%)	Yes	33 833 (36.73)	6070 (41.37)	6180 (38.89)	6666 (35.85)	6939 (34.43)	7978 (35.00)	0.13 (0.11 ; 0.15)
Therapy before index stay, n (%)	No therapy	26 289 (28.54)	4257 (29.01)	4509 (28.37)	5407 (29.08)	5758 (28.57)	6358 (27.89)	0.02 (0 ; 0.05)
	Monotherapy	35 981 (39.06)	5653 (38.53)	6283 (39.54)	7219 (38.82)	7878 (39.09)	8948 (39.26)	-0.01 (-0.04 ; 0.01)
	Double therapy	24 051 (26.11)	3846 (26.21)	4103 (25.82)	4824 (25.94)	5273 (26.16)	6005 (26.34)	0 (-0.02 ; 0.02)
	Triple therapy	5790 (6.29)	917 (6.25)	997 (6.27)	1146 (6.16)	1247 (6.19)	1483 (6.51)	-0.01 (-0.03 ; 0.01)
Gini index 2018	Mean (SD)	0.35 (±0.06)	0.37 (±0.07)	0.34 (±0.05)	0.35 (±0.05)	0.34 (±0.05)	0.36 (±0.05)	0.17 (0.14 ; 0.19)
	Median (Q1;Q3)	0.35 (0.31 ; 0.39)	0.36 (0.32 ; 0.41)	0.33 (0.30 ; 0.37)	0.34 (0.30 ; 0.39)	0.34 (0.30 ; 0.38)	0.35 (0.32 ; 0.39)	

CI, confidence interval; GP, general practitioner; HF, heart failure; LTD, long-term disease; Q1–Q3, first and third quartiles; n (%), number (and percentage); Q1–Q5, quintiles of the deprivation index; Q1, least disadvantaged; Q5, most disadvantaged; SD, standard deviation. Effect sizes were calculated between Q1 and Q5 only. Cohen's d was calculated for quantitative variables, and Cohen's h was calculated for qualitative variables. Cohen's d and h effect sizes are interpreted as 0.2: small, 0.5: medium, and 0.8: large.

Table 2 Outcome

Variable	Modality	Population n = 92 111	Deprivation index quintile					Effect size (Cohen's h) and 95% CI between Q1 and Q5
			Q1, n = 14 673 (16)	Q2, n = 15 892 (17)	Q3, n = 18 596 (20)	Q4, n = 20 156 (22)	Q5, n = 22 794 (25)	
Readmission within 1 year of follow-up, n (%)	Yes	27 312 (29.65)	4336 (29.55)	4579 (28.81)	5509 (29.62)	5964 (29.59)	6924 (30.38)	−0.02 (−0.04 ; 0)
Death within 1 year of follow-up, n (%)	Yes	23 905 (25.95)	3688 (25.13)	4205 (26.46)	4838 (26.02)	5288 (26.24)	5886 (25.82)	−0.02 (−0.04 ; 0)
Readmission within 2 years of follow-up, n (%)	Yes	31 170 (33.84)	4946 (33.71)	5204 (32.75)	6254 (33.63)	6828 (33.88)	7938 (34.82)	−0.02 (−0.04 ; 0)
Death within 2 years of follow-up, n (%)	Yes	36 506 (39.63)	5668 (38.63)	6340 (39.89)	7352 (39.54)	8071 (40.04)	9075 (39.81)	−0.02 (−0.04 ; 0)

CI, confidence interval, n (%), number (and percentage); Q1–Q5, quintiles of the deprivation index, Q1, least disadvantaged; Q5, most disadvantaged; Cohen's h effect sizes are interpreted as 0.2: small, 0.5: medium, and 0.8: large. Effect sizes were calculated between Q1 and Q5 only. Cohen's d was calculated for qualitative variables, and Cohen's h was calculated for quantitative variables. Cohen's d and h effect sizes are interpreted as 0.2: small, 0.5: medium, and 0.8: large.

system has gaps in prevention,^{29,30} it is effective in managing patients at the acute stage, due to relatively equitable access to care, regardless of economic barriers.

Disparities in the literature highlight the varying effectiveness of healthcare systems in addressing the needs of different populations. For instance, countries like France prioritize universal healthcare access to promote equitable medical services and mitigate these inequalities,³¹ while in the USA, disparities in healthcare access and outcomes persist, often exacerbated by socioeconomic .²⁻³⁴ In the context of HF, few studies conducted in the USA indicate that SES plays a critical role in determining health outcomes, with disadvantaged populations experiencing higher rates of mortality and morbidity due to barriers in accessing timely and appropriate healthcare services.³⁵⁻³⁷ Specifically regarding one-year mortality post-hospitalization for HF, research in the USA typically reveals a greater association than what we observed, with 36% increased risk of one-year mortality.³⁸ North America also have the highest one-year readmission rates among all regions, recorded at 63%, nearly double the rate reported in our study.^{39,40} Notably, socioeconomic disparities in healthcare access significantly diminished under the Affordable Care Act (ACA).⁴¹ Additionally, findings from countries with universal healthcare tend to reflect trends similar to those in our study⁴² or indicate no significant differences in mortality rates.^{43,44} In Spain, one study similarly suggests that although the gap remains significant, overall mortality differences may be less pronounced in universal systems.⁴³ Even within the NHS, which provides universal coverage, a deprivation gap in survival persists.^{45,46} Findings can yield mixed results depending on the socioeconomic indicators used.²⁰

The similar HR across all quintiles compared to Q1 suggests that more privileged individuals have a lower risk of death. This suggests that the advantage is concentrated among the most privileged group rather than following a linear gradient across all deprivation levels. This pattern may reflect better access to resources, healthier living conditions, and greater continuity of care in the least deprived areas. They may also be more likely to access preventive care in a timely manner, maintain more regular contact with healthcare providers, experience fewer delays in seeking medical attention, and show higher adherence to recommended treatments.⁴⁷ In contrast, among the other groups, the risk of death appears to be relatively uniform. This indicates that these individuals face comparable mortality risks, regardless of their specific SES. Once a certain threshold of socioeconomic disadvantage is reached, additional deprivation may have limited incremental impact. This pattern highlights a significant divide between the most privileged and the rest of the population, rather than a linear relationship.

Association with income inequality

The GINI index, which assesses income inequality within a city, did not reveal a significant correlation with overall mortality or readmission rates after hospitalization. This indicates that income inequality in a city does not seem to affect the likelihood of HF patient's mortality or being readmitted after discharge.

Strengths and limitations

Some limitations should be considered. This study was based on observational data, limiting the ability to establish causality. Potential confounders such as smoking status, as well as key clinical variables including left ventricular ejection fraction, New York Heart Association class, and measured proBNP levels, were not available in the SNDS and may have influenced the results. As the study population was limited to French patients previously hospitalized for acute HF, the results may not

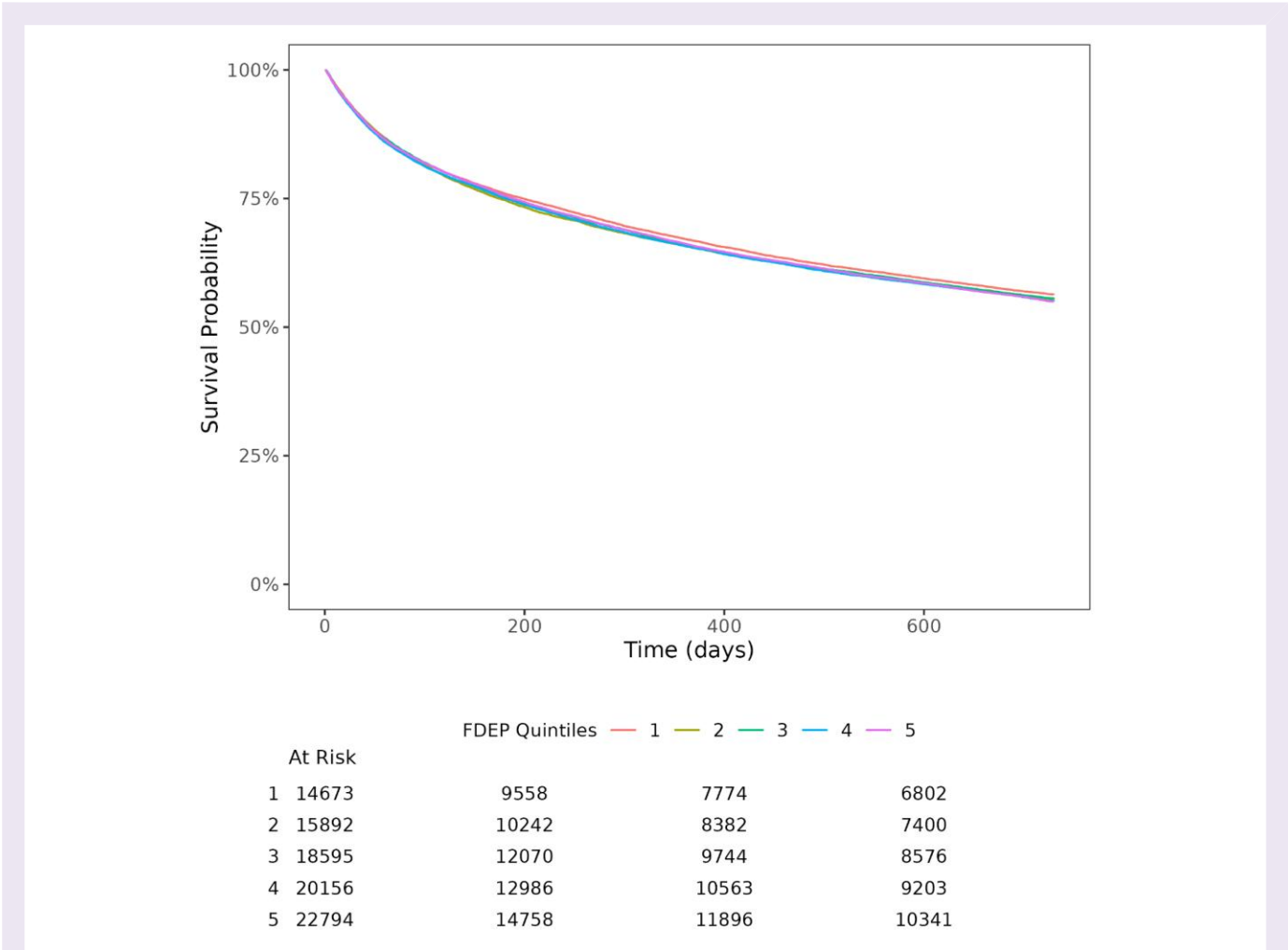


Figure 1 Overall survival by deprivation quintiles.

Table 3 Results for all-cause mortality at 2 years

Variable	Modality	Two-year all-cause mortality model					
		Crude HR	95% CI	P-value	Adjusted HR	95% CI	P-value
Deprivation index quintile	Q1: least disadvantaged	Reference			Reference		
	Q2	1.05	1.0–1.08	0.01	1.07	1.03–1.11	0.0002
	Q3	1.03	0.99–1.07	0.10	1.05	1.02–1.09	0.005
	Q4	1.05	1.01–1.08	0.01	1.09	1.05–1.13	<0.0001
	Q5: most disadvantaged	1.03	1.0–1.08	0.04	1.11	1.07–1.15	<0.0001
Gini index		0.94	0.75–1.08	0.26	0.94	0.78–1.13	0.49

Results are adjusted on readmission, age group (18–64, 65–74, 75–84, 85–94, 95+), sex, Gini index, establishment status, length of stay in hospital, transfer to intensive care, HF stability score, comorbidity [ischemic heart diseases, cerebrovascular diseases, peripheral vascular diseases, diabetes, cancer, dementia (including Alzheimer's disease), chronic respiratory diseases (including asthma and COPD), end-stage renal diseases], number of visits with a GP, the number of nursing care sessions, at least one visit with a cardiologist, the number of proBNP tests last year, and type of therapy in the last year.

Global univariable and multivariable P-values of the deprivation index quintile were 0.08 and < 0.0001, respectively.

CI, confidence interval; GP, general practitioner; HF, heart failure; HR, hazard ratio.

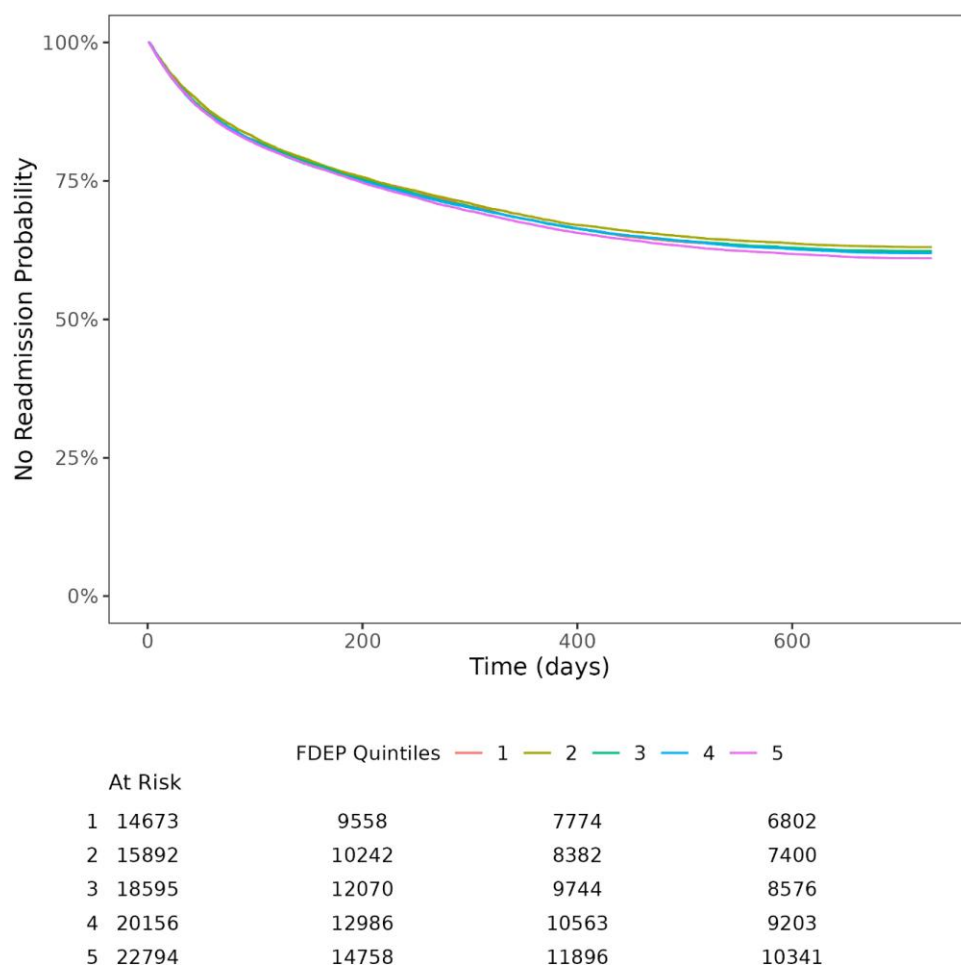


Figure 2 Readmission by deprivation quintiles.

Table 4 Results for HF readmission at 2 years

Variable	Modality	Two-year HF readmission model					
		Crude HR	95% CI	P-value	Adjusted HR	95% CI	P-value
Deprivation index quintile	Q1: least disadvantaged	Reference			Reference		
	Q2	0.97	0.94–1.01	0.15	0.97	0.94–1.01	0.20
	Q3	1.00	0.96–1.04	0.92	1.00	0.96–1.04	0.98
	Q4	1.01	0.97–1.05	0.60	1.01	0.97–1.05	0.60
	Q5: most disadvantaged	1.04	1.00–1.08	0.05	1.04	1.00–1.07	0.06
Gini index		1.11	0.92–1.35	0.28	1.03	0.84–1.25	0.8

Results are adjusted on readmission, age group (18–64, 65–74, 75–84, 85–94, 95+), sex, Gini index, establishment status, length of stay in hospital, transfer to intensive care, HF stability score, comorbidity [ischemic heart diseases, cerebrovascular diseases, peripheral vascular diseases, diabetes, cancer, dementia (including Alzheimer's disease), chronic respiratory diseases (including asthma and COPD), end-stage renal diseases], number of visits with a GP, the number of nursing care sessions, at least one visit with a cardiologist, the number of proBNP tests last year, and type of therapy in the last year.

Global univariable and multivariable P-values of the deprivation index quintile were 0.007 and 0.014, respectively.

CI, confidence interval; GP, general practitioner; HF, heart failure; HR, hazard ratio.

translate directly to other populations, healthcare models, or younger patients with less severe condition. The study included only patients admitted from and discharged to their residence, which increases population homogeneity but may limit generalizability. Another limitation lies in the fact that socioeconomic deprivation was observed at the area level. The findings therefore reflect outcomes for patients residing in more deprived areas rather than outcomes at the individual level of socioeconomic deprivation. Area-level variables are generally less accurate and may obscure associations. On the other hand, a recent meta-analysis found no clear trend showing that single vs. composite indicators or area vs. individual-level indicators were more effective in predicting outcomes.²⁰ The French deprivation index is based on four dimensions (income, education, employment, and unemployment) and does not account for other factors included in indices such as the UK Index of Multiple Deprivation, notably housing conditions, access to services, or environmental characteristics. These missing domains may affect health outcomes and could lead to residual confounding, which may contribute to the modest association observed between deprivation and mortality in our study. Several deprivation indices exist internationally. Comparative studies using alternative definitions are needed to determine whether other indices capture socioeconomic disparities more accurately than the French index and to assess the extent to which its structure may limit the detection of such gradients in HF outcomes. Since our measure of socioeconomic deprivation relies on area-level indices, the observed association may also be subject to the ecological fallacy, and caution should be exercised when interpreting these findings at the individual patient level. The lack of association with the Gini index may also reflect the limited relevance of income inequality measured at the municipal level, which may not capture broader economic patterns affecting health. Additionally, cause-specific mortality was not used because it is often unreliable in large epidemiological studies relying on death certificates, where diagnostic accuracy and coding consistency are limited.⁴⁸

This study has also several strengths that enhance its validity and relevance. It utilizes an exhaustive, nationwide cohort of over 92 000 patients hospitalized for HF in France, providing a robust sample size that increases the reliability of the findings. The comprehensive data collection from the national health insurance programme allows for detailed analysis of sociodemographic factors, adjustment for numerous comorbidities, and healthcare utilization.

Conclusion

Within the French universal healthcare system, we found no major differences in HF readmission risk across socioeconomic strata, and the Gini index was not associated with either mortality or readmission. The only notable difference concerned mortality, which was modestly higher in all but the least deprived quintile. These findings suggest that, unlike in the USA where socioeconomic gradients are more pronounced, France's universal healthcare system may mitigate socioeconomic disparities in outcomes after HF hospitalization.

Author contributions

GM and FR supervised and conceptualized the study. All authors contributed to the study methodology. VG performed the data extraction and statistical analyses. HG drafted the manuscript. All authors contributed to the study, critically revised the manuscript, reviewed, and approved the final version.

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Conflict of interest: HG and VG declare that they have no conflicts of interest. GM declares research grants paid to the University Hospital Center (CHU) of Montpellier by the Novartis, as well as paid participation on the Safety Monitoring Board of Roche and stock ownership as a minority shareholder in the company KanopyMed. G.B. reports personal fees from Abbott, AstraZeneca, Boehringer Ingelheim, Novartis, and Novo Nordisk, outside the submitted work. HVS declares that she has received research grants from Boehringer Ingelheim Pharmaceuticals, Inc. (institutional educational account), the Canadian Institutes of Health Research, the Heart and Stroke Foundation of Canada, and Novartis Corporation (institutional educational account). She has served as a consultant for AstraZeneca, the Baim Institute for Clinical Research, Bayer, Medtronic, Novo Nordisk, and Roche. NG declares that he has received consulting fees from AstraZeneca, Novartis, Bayer, Novo Nordisk, Boehringer, Roche Diagnostics, Lilly, Echoscens, Cardiostory, and NP Medical. He has also received travel support from AstraZeneca, Novartis, Bayer, Novo Nordisk, Boehringer, NP Medical, and Lilly. FR declares honoraria for lectures from AstraZeneca, GSK, Servier, Boehringer, AstraZeneca, Vifor, Bayer, Pfizer, Novartis, Servier, Novo Nordisk, Air Liquide, Abbott, QuidelOrtho, Newcard, MSD, BMS, Sanofi, and Alnylam. He also declares grants from Air Liquide Healthcare and Abbott Medical, consulting fees from Abbott, Air Liquide, Bayer, and Pfizer, support for attending meetings from Novartis and Boehringer Ingelheim, participation on the Advisory Board of Carmat, leadership or fiduciary role in Boehringer Ingelheim, Vifor Pharma, and Novartis, and compensation for medical writing services provided from Pfizer.

Data availability

Data sets generated and/or analysed during the current study are not available to the public to protect the confidentiality of study participants.

Ethical approval

The methods were performed in accordance with current guidelines and regulations. The investigation conforms with the principles outlined in the Declaration of Helsinki. The study has been approved by the Institutional Review Board of the University Hospital of Montpellier (IRB ID: 202201065, Approval Number: IRB-MTP_2022_03_202201065).

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